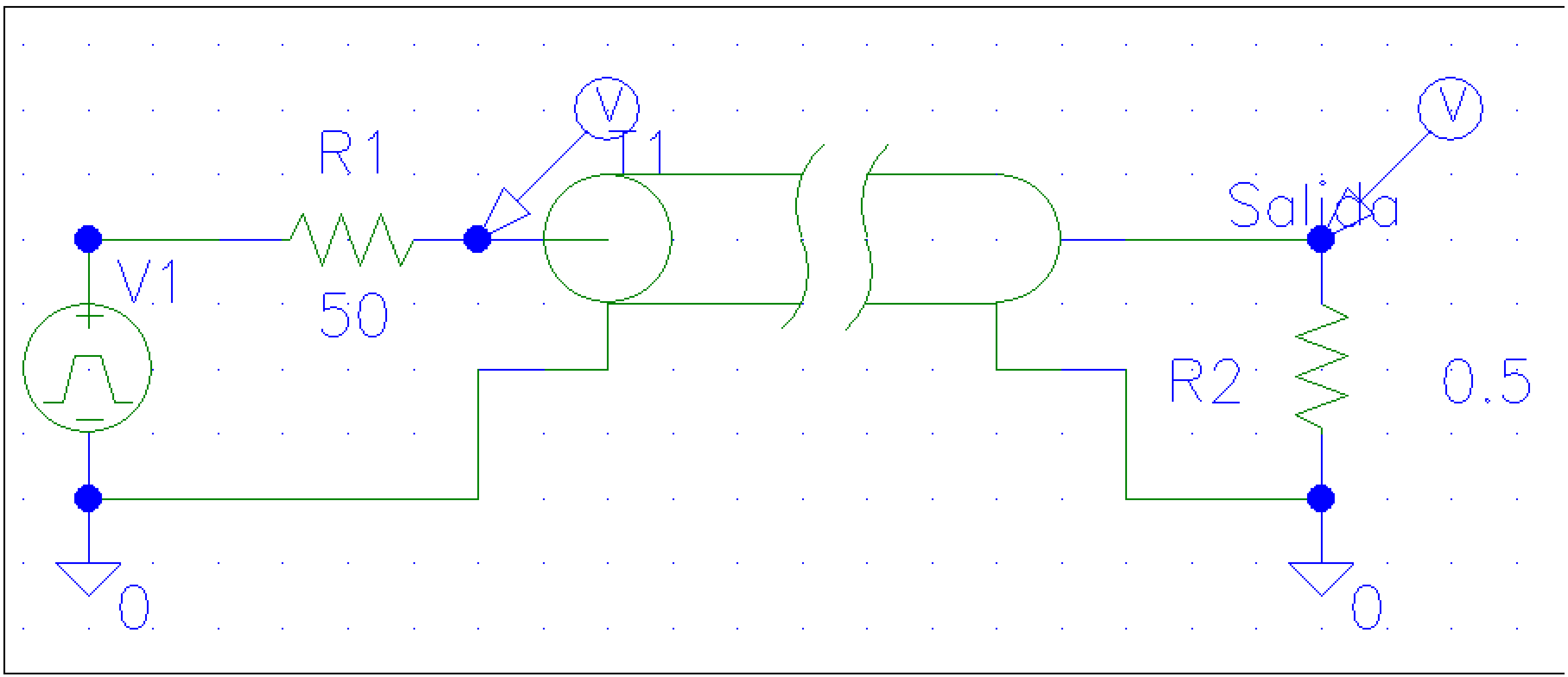
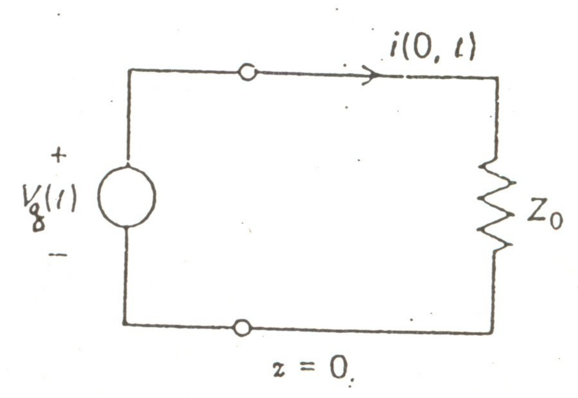
Práctica 4:

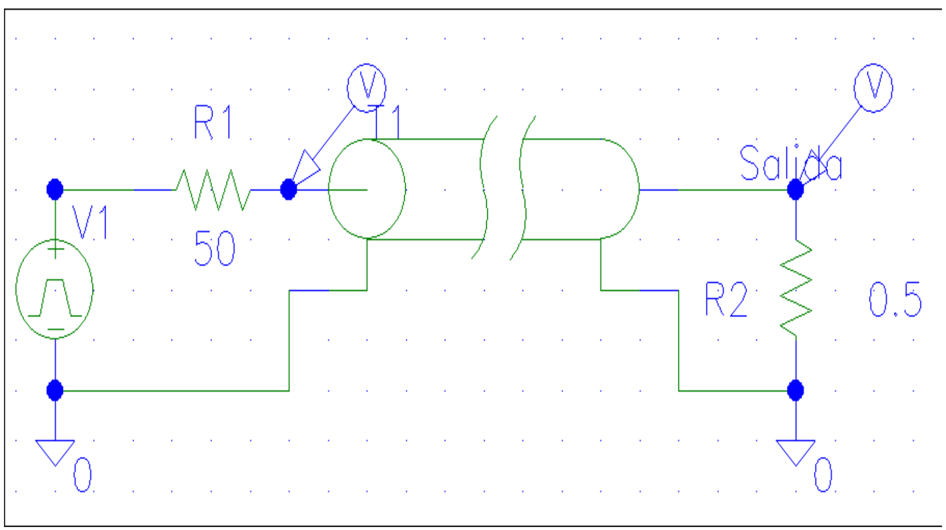
Simulación de líneas de transmisión

Circuito a simular con Pspice



Esquema eléctrico: cálculos





1 – Hallar Intensidad en T=0, z=0

$i(z=0,\ t=0) = i_0 = V/R = V1/(R1+Z_0)$

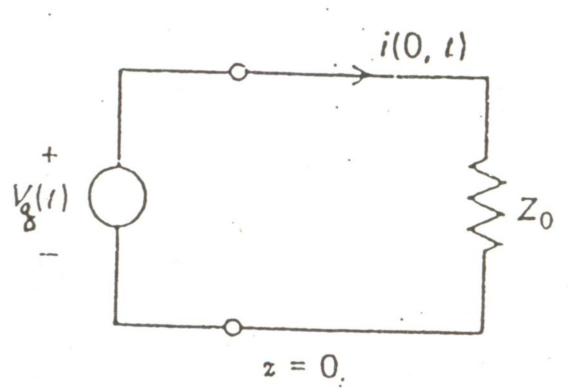
$i_0 = V1/(R1+Z_0) = 5/(50+75) = 0.04\ A$

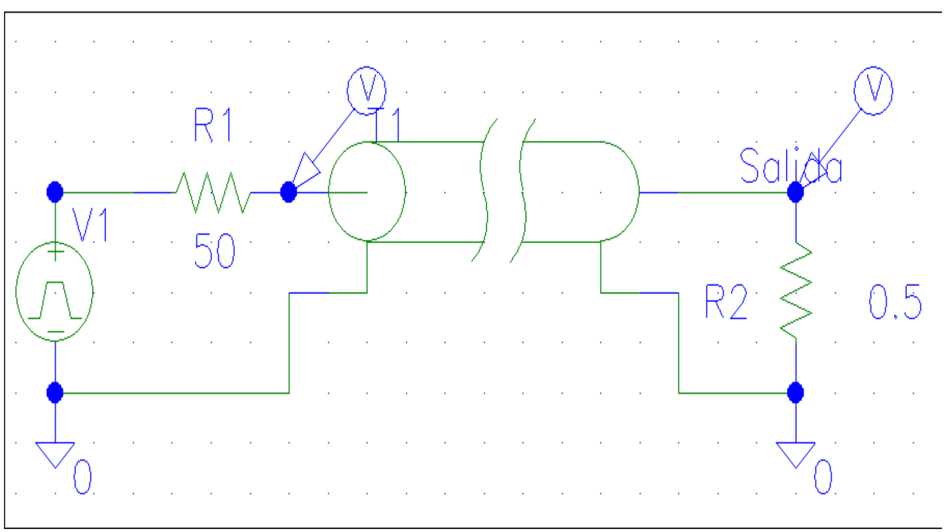
2 – Hallar tensión en T=0, z=0

$v(z=0,\ t=0) = V1 - i_0 * R1$

$v(z=0,\ t=0) = 5 - 0.04*50 = 3\ V$

Esquema eléctrico: cálculos





3 – Hallar V en Tp, z=l)

$v(z=l,\ t=Tp) = V_+ + V_- = 3 - 3 = 0$ V.

CRef. Carga $= (R2-Z_0)/(R2+Z_0) = V_-/V_+$

$v(z=l,\ t=Tp) = 3+V_- =$

CRef.$=(0.5-50)/(50+0.5)= -1 = V_-/V_+$

$V_- = -1 * V_+ = -3$ V.

4 – Hallar tensión en 2 Tp, 3Tp,...

$v(z=0,\ t=2Tp) = V_+ + V_- + V_{gen} =$

CRef. Gen. $= (R1-Z_0)/(R1+Z_0) = V_-/V_+$

$V_{gen}= 0$ V. (PER=1 ms., 2Tp= 800 ns.)

CRef.$=(50-50)/(50+50)= 0 = V_-/V_+$

$V_- = 0 * V_+ = 0$ V.

$V_+ = V_-$ de $v(z=l,\ t=Tp) = -3$ V.

$v(z=0,\ t=2Tp) = -3 + 0 + 0 = -3$ V.